Raffaele Errichiello

Architect, designer, researcher
Material Practices Lab
Umeå School of Architecture (UMA)
Umeå University, Sweden

raffaele.errichiello@umu.se
ORCID 0009-0005-6327-3641
raffaeleerrichiello.com, LinkedIn

I am an architect working with living materials. I combine computational design and digital fabrication with fungal biology to make building components that are 3D printed and then grown. Since my master's thesis in 2020 I have worked with mycelium in research, teaching and commissioned projects, from furniture and installations to printed and grown demonstrators.



Doctoral research

2024 – present

Geometry as a Design Parameter for Living Materials: Fungal Development and Structural Behavior in 3D-Printed Biochar-Mycelium Composites
Doctoral thesis (working title), Material Practices Lab, UMA. Supervisors Julio C. Diarte Almada (principal) and Gireesh Nair (co-supervisor, Department of Applied Physics and Electronics, Umeå University).

Education

2024 – present — **PhD in Architecture**, Umeå School of Architecture, Umeå University

2014 – 2022 — **Master's degree in Architecture (five-year integrated M.Arch, 300 ECTS)**, University of Naples Federico II, Italy
Thesis with Fabiana Polichetti, supervised by Prof. Sergio Pone: “Coltivare strutture. Bio-compositi e strutture parametriche” (2022), researched at UMA during an Erasmus+ exchange, 2019 to 2020. Qualified architect in Italy (Esame di Stato).

Positions

2024 – present — **Doctoral candidate in Architecture**, UMA, Umeå University

2024 — **Project assistant**, UMA, Umeå University

2022 – 2024 — **Guest teacher, Studio 12, master's programme**, UMA, Umeå University
Seminars and workshops on biomaterials, digital tools and digital fabrication.

2022 – 2024 — **Research assistant**, UmArts, Umeå University
Winter Garden and Pallet Parliament (2022), Nordic-Arctic Atlas (2023 to 2024).

Jan – Jun 2023 — **Guest researcher**, RISE Research Institutes of Sweden, Umeå
First printed tests of a mycelium-based, fibre-reinforced material; grew the Design S 2023 trophies with Ellinor Werner.

Apr – Oct 2022 — **Erasmus+ trainee in bio-based 3D printing**, UMA, with RISE Research Institutes of Sweden

2021 — **Junior architect**, FRAME, Naples

Publications

2026, Journal article — Errichiello, R., & Diarte Almada, J. C. **Experimental study of 3D printed biochar-mycelium composites via liquid deposition modeling for sustainable building components: compatibility, shrinkage control, structural integrity and prefabrication.** *Architectural Intelligence*, 5, 16. doi.org/10.1007/s44223-026-00125-7

2026, Book chapter — Errichiello, R., & Diarte Almada, J. **Biobased Material Confluences: Investigating Material Formulation and 3D Printing Parameters for Biochar-Mycelium Building Composites.** In Y. Liu, H. Chai, D. W. Bao & P. F. Yuan (Eds.), *Transindividual Intelligence* (CDRF 2025), pp. 297–307. Springer. doi.org/10.1007/978-981-92-0615-5_26

2025, Journal article — Errichiello, R. **Anatomy of a Living Joint: From Hyphal Networks to Urban Fabrics.** *UOU Scientific Journal*, 10, 128–135. doi.org/10.14198/UOU.2025.10.13

2024, Conference paper — Errichiello, R. **AURORA: Adaptive Units for Rational, Organic and Resilient Architecture. Form Finding of Biomimetic Minimal Surfaces.** Proceedings of *SIGraDi 2024*, Barcelona, pp. 2383–2394. doi.org/10.52842/conf.sigradi.2024.2383

Grants awarded

SEK	Funder and purpose
130,000	**MIRAI ECR mobility grant**, research stay at Tohoku University, Sendai
100,000	**UmArts Small Visionary Seeds**, “Grow, Join, Dry”, decay and degradation over one year
53,400	**Erasmus+**, three grants: SDU summer school, TU Delft course, eCAADe 2026
46,000	**Kempe Foundations**, three grants: fermentation chamber, Tokyo visit, CDRF 2025
5,500	**eCAADe Grant for Young Researchers**, eCAADe 2026, Lübeck
334,900	**Total**, 2025 to 2026

Built and grown work

	Project	Place	Role
2025	**Telescopic canoe** laminated-wood skin with double curvature	ProtoLAB, Wrocław	Tutor, with J. Sánchez Merina
2024	**Armadillo** timber arches with nests for insects	ProtoLAB, Wrocław	Tutor
2023 – 2024	**Letter N for Netflix** grown mycelium logo, 3D-printed brackets	Netflix, Stockholm	Commissioned by Open Studio Stockholm
2023	**Design S trophies** trophies grown from reishi mycelium	Svensk Form, Stockholm	Commissioned by RISE and Svensk Form
2022	**The Winter Garden** inflatable dome installation	Umeå Arts Campus	Form-finding, fabrication
2022	**Pallet Parliament** amphitheatre of reused euro pallets	Teg, Umeå	With A. Haiek Coll, R. Rudolph
2022	**MK3** bent steel tubes and rope nets	UMA, Umeå	Design, fabrication
2022	**REishi chair** mycelium seat on turned wooden legs	UMA Furniture Fair	Designed and grown; awarded
2020	**Stasys** fabric shell prototype in the snow	UMA, Umeå	With F. Polichetti
2016	**haMMMocks** three hammocks on three chestnut poles	Lecce, Italy	With fellow students

Exhibitions

Jun 2026	Demonstrators D1 and D2, EURAU26 LATITUDES	Umeå
Apr 2024	REishi chair in NON-SITE, UMA Furniture Fair	Umeå
Jan 2024	Nordic-Arctic Atlas of Environmental Injustices, UmArts	Umeå
2023	“Living Architecture”, first printed mycelium pieces, RISE	Umeå

Talks

Oct 2026	UmArts Research Week, talk “Geometry as a Design Parameter for Living Materials”	Umeå
Sep 2026	eCAADe 2026, PhD workshop, thesis presentation	Lübeck
Jun 2026	EURAU26 LATITUDES, paper “From Biomass to Building”	Umeå
Dec 2025	6th World CAAD PhD Workshop, as eCAADe representative	Online
Jul 2025	CDRF 2025, paper presentation, Tongji University	Shanghai
Jul 2025	ProtoLAB 2025, research presentation	Wrocław
Jun 2025	TU Delft, PhD course, talk “Shaping Forces, Growing Forms”	Delft
Nov 2024	SIGraDi 2024, paper presentation	Barcelona
Feb 2024	UMA Research Day, talk “Growing Architecture”	Umeå

Teaching

2022 – 2024	Workshops “Imaginative Scenarios” and “Rhino to AI”, Studio 12	Umeå
2022 – 2024	3D-printing sessions, Studio 12	Umeå
2022 – 2024	Seminars on mycelium substrates, with lab sessions at RISE	Umeå

Service and collaborations

2026 – present	PhD representative on the PhD and Research Committees, UMA	Umeå
2026	EURAU26 LATITUDES, session chair and abstract reviewer	Umeå
2026 – present	Research collaboration, Forest Ecology	Tohoku University
2025 – present	Research collaboration, Department of Molecular Biology	Umeå University
2025 – present	Member, COST Action Designae	Europe

Awards

2022	Award for the REishi chair, first UMA Furniture Fair	Umeå
2018 – 2019	First prizes in two categories, Pattern Your Life competition, VAHA	Naples

Skills and languages

Fabrication	Robotic LDM printing (UR15), FDM and SLA printing, laser cutting, CNC milling
Materials lab	Mycelium cultivation (*Ganoderma lucidum*), sterile technique, compression testing
Computation	Rhinoceros 3D, Grasshopper (Kangaroo2, Karamba3D, Galapagos), Python, C#, Git
Analysis	Life-cycle assessment (One Click LCA), Ladybug, Honeybee, R, ImageJ/Fiji, OpenCV
Drawing	AutoCAD, Archicad, Revit, Blender, V-Ray, Enscape, Adobe Creative Cloud
Languages	Italian (native), English (fluent), Spanish (intermediate), Swedish (basic)